

Why you should give your Students Automatic Process Feedback on their Collaboration: Evidence from a Randomized Experiment.

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Abstract In Computer-Supported Collaborative Learning (CSCL), students learn in small groups to achieve learning benefits outside what would be possible for individual students. As in other forms of learning, students need feedback on the quality of their work. Still, writing high-quality informative feedback requires time on the part of educators. This makes it less feasible to provide quality feedback at scale. Given the importance of group dynamics in CSCL, quality feedback should also contain information about group processes and discussion quality. Emergent roles provide a natural anchor point for this.

We propose a method to automatically deliver highly informative, textual process feedback for CSCL forum discussion tasks in an introductory teacher education class, focusing on individual students' group communication. This feedback is generated from several discourse indicators, that are used to derive emergent roles of the students. From that we derive feedback about strengths and potential for improvement. In a randomized control trial, we show that the highly informative feedback generated with our method is preferred by the students. Implications and avenues for future research are discussed in light of these findings.

Keywords: Automated Textual Feedback · AI-assisted Agents
· Collaborative Learning · Learning Analytics

1 Introduction

In higher education, researchers and educators often find themselves in a position where high student-to-teacher ratios make it difficult to provide personalized support to students. There are two main ways to resolve this: (1) Students can learn with their peers in social learning environments. (2) Teaching personnel can

be supported in reaching more students at once. To combine these approaches, methods from Computer-Supported Collaborative Learning (CSCL) can be accompanied by Learning Analytics (LA), leading to cycles of social knowledge building and data-informed self-reflection. To move from LA towards informing learners' self-reflection, LA data must be interpreted and brought into a form that learners can understand, like well-phrased textual feedback [23]. By pre-selecting and contextualizing data, textual feedback can overcome the drawbacks of other forms of data display like learning dashboards, which often only visualize learning data without proper context [18]. Showing relevant information with an interpretation makes such feedback highly informative [11] and helps students to become competent agents within their reflection processes.

Providing students with feedback on their writing and discourse structure during and after their discussion assignments bears the potential to support them in reflecting on their collaboration quality. Ideally, this would improve their metacognition in communicative situations and thus help to make their learning groups more effective. Our goal is thus to advance the methods for automatically providing feedback to students (1) *by evaluating the use of mostly linguistic measures (indicator candidates) to assess qualities of educational forum discourse* and (2) *by providing highly informative feedback based on language and communication analysis*. We apply various means of discourse analysis, including readability assessment, psycholinguistic measurements, and group communication analysis to a data set of educational discourse. The data set was collected during a CSCL task in a teacher education lecture. The assignments were specifically designed for a project that aimed to provide automatically generated feedback to students. Finally, we provide preliminary evidence from a randomized control study that our feedback is liked by students both in comparison to less informative feedback and in absolute terms.

2 Background

CSCL as a discipline often draws from the *socio-constructivist framework* [15, p. 17], which also serves as a basis for the research at hand. Constructivists argue that knowledge is built actively and all knowledge is constructed in a cycle of acting (construction), perceiving (reconstruction) and reflecting (deconstruction). In socio-constructivism, this idea is taken one step further by arguing that the knowledge construction process developing the human intellect mostly takes place in social environments and is therefore heavily influenced by those [e.g., 9]. This is supported by psychological findings that all parts of the knowledge construction process are influenced by social environments [e.g., 7, 12]. Social environments and the emerging role constellations that prevail within them hence play a decisive role in the quality of learning outcomes. Emergent (social) roles therefore provide a natural anchor point when aiming to support knowledge construction processes. Dowell, Nixon, and Graesser [10] operationalize the identification of emergent roles on a linguistic level through their Group Communication Analysis method and provide evidence that supports the theory.

In CSCL, educators create social contexts using technology such as discussion forums. They do this by integrating the knowledge construction cycle phases into CSCL apps, with assignments following a macro script of learning activities, which in turn follow a micro script [8]. To aid learners in cognitive regulation between phases, technology can either mediate human intervention or intervene automatically. As group dynamics are so important in CSCL, *Data-enriched Learning Activities* [11] for CSCL should stimulate the learners' metacognition, enabling them to reflect on their group role and to self-regulate if required.

To achieve this, we can give process feedback to learners. Data-informed, personalized, and textual feedback for learners brings Learning Analytics and socio-constructivist thinking together by creating a formative learning environment [27], moving us further towards *Highly Informative LA* [11]. Such feedback should be concise, actionable [4], and in line with learners' goals [18]. The automatic observation of collaborative knowledge construction and deconstruction can be achieved by analyzing group communication patterns and identifying emergent roles. In any CSCL task, role identification reveals profiles of learners, with their communicative strengths and weaknesses. This knowledge helps us to focus on important information for each role and tailor feedback accordingly. Such highly informative process feedback can then inform learner reflections.

3 Research Questions and Methodology

For the present research, we collected forum posts and forum metadata from a CSCL discussion assignment. We collected the data from a first cohort of students, so that we could then analyze it before running same assignment again with another cohort. Our goal was to automatically create LA-informed feedback for the second cohort of students using the insights that we gained from the analysis of the first cohort's data. Our analysis in this paper is mainly based on linguistic process indicators derived from the forum messages in the data set and their interrelations.

The research questions that we want to answer in the present research are:

RQ1a What are useful indicators for analyzing CSCL discussion assignments?

RQ1b Using a select set of indicators, what emergent roles can we identify in the data set?

RQ2a How can the examined indicators and emergent roles inform personalized, textual process feedback in the context of the given assignments?

RQ2b Do students find the provided feedback useful to reflect and self-regulate?

We describe each indicator candidate in detail and provide a rationale for its potential use in an educational discourse analysis. Based on normative, as well as evidence-supported argumentation, we select the indicators that we want to use for our emergent role analysis and in our feedback texts (RQ1a). To assess how each selected indicator should influence the feedback texts (RQ2a), we test the indicators for the stability of their correlations over the assignment phases. Thereafter, to detect emergent roles (RQ1b), we carry out a cluster analysis.

Indicator candidates that had significant impact on the formation of the clusters will be singled out for phrasing personalized feedback (RQ2a).

We then suggest feedback texts for each cluster. Some indicators might only be relevant for certain groups of students, for example depending on the cluster they belong to. They will thus be included into the feedback conditionally. In a randomized control trial, we gave feedback texts generated using this method to a new cohort of students. Those texts also contained generated product feedback on the group outcome. We show that the highly informative feedback generated with our new method is preferred by the students (RQ2b). Implications and avenues for future research are discussed in light of these findings.

4 The CSCL Discourse Data Set

The data set under examination was created within one of our research projects. The project aims towards analyzing the manifold data that is generated during the course of online learning to give automated, artificial intelligence (AI) based feedback to learners.

Part of this project was a forum discourse assignment, where first semester student teachers had to discuss two videos of teaching lessons in three assignment phases. 689 students contributed 8115 posts in total to the discourse forum. The students were put into groups with three up to six students, most of which had four (31 groups) or five (144 groups) participants. This resulted in a total count of 187 groups. The assignment was split into three phases⁴:

1. In phase 1, the students had to watch the two videos and then collect two to six teaching activities (group work, class discussion, etc.) for each video. Every student had to contribute at least one activity here and contribute at least one additional comment to the forum. (2849 posts)
2. Next, in phase 2, the students had to analyze the collected activities regarding their potential for cognitive activation of the students shown in the video. Members of the group would then either agree to the analysis of their peers or provide their own alternative analysis. (3861 posts)
3. In the last phase, the students were to formulate a collective statement based upon their analyzes from phase two. One or more students would draft a statement and the others would then comment their improvements on that statement, until every student agreed on the final solution. (1405 posts)

All three phases had a strong script on how to use the forum. The students had to place each statement in a separate comment. Also, to express affirmation, students had to use (in German) the phrases “this statement reflects my opinion” and “I agree with the assessment”.

⁴ The assignment texts, as well as other supplemental material, like statistics about the data, the database schema, formulas, algorithms, our evaluation data, and the feedback templates can be found in our OSF.io repository.

5 The Indicator Candidates

5.1 Group Communication Analysis

Group Communication Analysis (GCA) is a method that combines computational linguistic analysis techniques with time series analysis⁵. It was invented by Dowell, Nixon, and Graesser [10], who used it to detect emergent social roles in two large data sets of educational chat discussions. The method combines several measures which are mainly based on semantic vectors and vector space calculations with a post count based participation measure.

We made two changes to the original method: first, we used Sentence BERT [24] instead of Latent Semantic Analysis (LSA) to generate our vectors. This allowed us to skip the corpus expansion and model training steps. Also, output vectors from large neural models like Sentence BERT are known to yield better results for semantic analyses, than LSA does. Secondly, the original GCA method worked for chat data, which has no additional information about the relations of posts, other than the order of their arrival. Our forum data was structured as a post-reply-tree. We took this into account by only comparing posts that had a child or sibling relation within that tree⁶.

Participation The participation measure is there to “signif[y] a willingness and readiness for participants to externalize and share information and thoughts” [10, p. 1010, 3]. The measure sets the post count of a participant in relation to the post count of the discussion. It rewards participants with a high post count share relative to the whole conversation. We used the improved participation measure that the authors supply [10, fn. 1].

Responsivity The internal cohesion, responsivity and social impact all describe how participants interact with each other and with themselves on a semantic level. The responsivity of a participant signifies how they reflect on other participant’s contributions [e.g. 16]. It is measured by comparing the content of all posts a participant made to all previous posts by the other participants.

Social Impact The social impact then is the flip side of a participant’s responsivity: it shows how other’s reflected on a participant’s posts by comparing a post to all posts that were made later in time by other participants. The responsivity and the social impact are therefore a quantification method for collective knowledge construction.

Internal Cohesion The *internal cohesion* of a participants posts was constructed to quantify their reflection process and therefore self-regulation [e.g. 30]. It works by comparing the content of each post of a participant to all previous posts that a participant made. The measure’s value rises in proportion to the semantic similarity of the posts of an individual.

⁵ you can find all the formulas we used in the supplemental material but if you are interested in the topic, go read the original paper. It’s really a masterpiece.

⁶ See the supplemental material for more information on how we transformed the graph.

Newness The newness quantifies the tendency of a participant to share new information, rather than to discuss mostly the information that has already been shared (*biased information sharing*) [6] and to pass over critique or alternatives to the current consensus [17]. It is measured by averaging over the distance of each of a participants posts to the semantic space of the previous posts in a discussion, which is represented by a linear hull.

Communication Density This measure describes how dense a participant communicates. With the changes we made to the original method, we couldn't implement this measure as originally described. We thus had to replace it later with the Flesch Reading Ease score. Still, in future research, we might try to replace Flesch with another measure that can capture the density on a semantic level.

5.2 Affirmation Detection

It was an integral part of all assignments that the students explicitly affirmed the statements of their group members. Affirming peer contributions was the central mechanism to keep the discussion vivid and to settle on statements and on assertions alike. To detect affirmation, we created semantic vectors for a list of common affirmative comments. We then took the average vector $\vec{a}$ of these and compared the vectors of all sentences in a post to $\vec{a}$. When a sentence vector had a cosine distance below a threshold of 0.461 (chosen by manually examining the data), the post was marked as containing an affirmation. The affirmation rate of a student is the share of their posts which contain an affirmation.

5.3 Readability

Evidence that the reading ease of a text plays an important role for a reader's engagement with its content exists for a very long time by now. Reading ease has therefore played a role in field of psycholinguistics for nearly a century by now [cf. 13]. We expected that the readability of a participants contributions in a text-based online discussion does influence the engagement of other participants with a contribution as well.

To measure readability, we implemented the Flesch Reading Ease score [13]. The parameters of the Flesch formula has to be recalculated for every language. We thus use the formula from Amstad [1] as an indicator candidate.

In section 5.1, we mentioned that we could not implement the *communication density* from the GCA. We still wanted a measure that served a similar purpose. Explaining complex matter using simple words usually makes texts less dense but more readable. Therefore, we deemed the Flesch score to represent an inverse of the idea behind the *communication density* and used it as a somewhat imperfect replacement measure. It is imperfect, because it doesn't capture the density on a semantic level, but only based on sentence and word lengths.

5.4 Uncertainty

For educational and scientific discourse, the presence of uncertainty poses a key feature. Chen, Benus, and Hernandez [5] even argue that science in general is about the management of uncertainty. They expound the problems of presenting merely the final result of science – a knowledge product – as being the science itself instead of acknowledging that science is mostly about the process of constructing knowledge. This process is about transitioning from an uncertain idea of how something could be – i.e., a hypothesis – to certain knowledge about the examined matter. While science is about using a methodology for constructing new, formerly “unknown knowledge”, education is about reconstructing “known knowledge”. In education, uncertainty therefore needs to be managed, too. Albeit implicitly, it has to play a key role in any well-designed didactic method. Observing uncertainty in discourses within science education settings is therefore an important factor for gaining an understanding of the discourse quality.

In natural language, there are certain cue words and phrases, but also semantic structures beyond those that signal hedging and semantic uncertainty [21]. Cues of uncertainty in language have highly contextual semantics. Hence, the cues have to be contextually disambiguated [32]. We used a custom transformer model based on RoBERTa [22] to detect four types of uncertainty: *epistemic*, *doxastic*, *investigative* and *conditional* [31]. For each student, we then used the mean rate of uncertainty tokens per post as an indicator.

5.5 Other Candidates

There were several indicator candidates which we also considered but which did not turn out to be too useful for us. To keep the cluster analysis to a manageable amount of indicators, we decided to rule out these indicators from the analysis. Still, we kept the option to use them in the feedback templates if they would turn out to be distinctive for a certain cluster *after* the cluster analysis.

Activity Based Collaboration Analysis This set of indicators, including *initiative*, *responsiveness*, *presence*, and *productivity*⁷, were derived by Scheffel *et al.* [29] from forum metadata and used to predict and influence student performance. While they serve as a baseline for comparison with the linguistic indicators in this paper, they are not included in the cluster analysis as they are not based on language properties. Additionally, the *participation* measure from the GCA method implements the same concepts.

Affective Norms Affective norms are psycholinguistic attributes of words. We speculated that the affective properties of communication would indicate the emotional involvement of a learner, which among other things plays a significant role for their social presence [26]. On the lexical level, using the dictionary from Köper and Schulte im Walde [20], we examined the *abstractness*, *imageability*, *valence* and *arousal* of the language used by each student.

⁷ as the students were in fixed groups, implementing the *connectedness* indicator [29] did not make sense for our data set.

Keyword Simple Word Rates Keywords are words that are of low frequency in common parlance but of higher frequency in the examined corpus. The keywords of our CSCL discourse corpus are primarily technical terms that relate to the assignment. We hence assumed that students’ use of keywords could indicate the level of confidence with which they navigate the field. Simple words are the flip side of this coin. By the our definition, these are words that occur at a comparatively high frequency in common parlance. Here, we used the 200 most common words in a subset of the German Colossal, Cleaned Common Crawl Corpus [25]⁸.

Mean post depth in the discussion graph For each student, we calculated the average relative depth of their posts in the discussion graph with $0 \leq \text{depth} \leq 1$. This would allow us to observe where the posts of a student concentrate in the discussion. We did not use this measure in the cluster analysis itself, though it turned out that this measure helps with explaining some of the emergent roles that we found later.

Length measures Lastly, we also measured the sentences per post, the ratio of accessory to main clauses and the sentence length in tokens.

6 Selecting and Evaluating the Indicators

To identify emergent roles using a cluster analysis (RQ1b), we had to make sure that we could assess the cluster composition. In addition, we needed to limit the effect of a potentially too similar set of variables to skew the cluster centroid distances. Students want concise, actionable feedback [4]. We can deliver this by concentrating only on those features that really matter. This meant that we now had to reduce the feature space by carefully selecting the indicators which we would use to determine the emergent roles.

Before we carried out any analysis, we normalized all the data by (1) win-sorizing each indicator candidate for their upper and lower percentiles, as well as (2) z-score-normalizing it. Both steps were more important for the cluster analysis than for the correlation analysis, as Spearman’s r_s can deal well with outliers and differing variable scales.

We then computed correlation matrices, using Spearman’s correlation coefficient for all indicators on the participant level and taking the mean values of all candidates that were computed on the post level. We generated one matrix for all discussions of a group, as well as one for each discussion phase. We also inspected the correlations between well-founded candidates and those that were not as well-founded to see if we could gain insights into the usefulness of the latter ones. But as we did not have information about how these candidates would influence the group outcome, this turned out to be a rather impractical approach. By combining the correlation analysis and a normative evaluation of the indicator candidates, in the end, we included the set of indicators into the cluster analysis that are described in sections 5.1 to 5.4.

⁸ The procedure to retrieve the simple words and the keywords is described in the supplemental material.

To answer RQ2a, we also needed to figure out how each indicator would influence the phrasing of our feedback. Each indicator candidate could potentially behave differently depending on the micro script of the assignment phase (RQ1a). Hence, we needed to assert the stability of the indicators over the phases. For unstable indicators, we will need to collect empirical data on their behavior with regard to a specific micro script in the long run. For the work at hand, we might have to assert them per phase in the feedback text, instead of just reporting, e.g., on an overall mean value. To assert the indicator stability, we computed paired t-tests over the raw correlation values between all indicator candidates and each phase. This produced a matrix for each of the candidates which showed the significance of the changes in an indicators correlations between each phase.

Only the affirmation rate and uncertainty indicators were unstable among phases. Uncertainty in phase 2 significantly differed ($p < 1\%$) from phases 1 and 3, likely due to the distinct tasks assigned in each phase: collecting information in phase 1, discussing in phase 2, and summarizing findings in phase 3. The affirmation rate also significantly varied ($p < 0.2\%$) between all phases, correlating with the amount of prompted interactions in each phase. This means that (1) the affirmation rate and uncertainty depend on the assignment’s micro script, and feedback on these indicators can’t be transferred to different micro scripts. Norms for these indicators would need recalculating for new micro scripts. Also, (2) derived clusters would be specific to these indicators. (3) Feedback given on these indicators should consider the given micro script.

7 Finding the Emergent Roles

The emergent roles (RQ1b) are inferred from the results of a hierarchical clustering analysis. Hierarchical (or agglomerative) clustering can support selecting an adequate cluster size visually, through dendrograms. This method helped us with selecting a feasible number of clusters by setting a minimum distance for the centroids and in naming the resulting roles by considering the proximity of the clusters. We set the cluster distance to 15.5, as we noticed that the last two clusters that emerge from that distance are very distinct. Unifying them by setting the distance just a little higher made the clusters more difficult to explain. We then made an independent t-test for each of the selected indicators comparing all clusters with each other, to determine which indicators are most distinctive between the clusters (sample groups). In the final written exam on the lecture, students were given a task on cognitive activation related to our CSCL discussion assignment. Their score on this task was included in our data set. As we wanted to know how the clusters relate to student performance, we also examined the relationship between clusters, exam points, and dropout rates.

Cluster 1 members contributed moderately but shared the most new and relevant information (as seen by their social impact), despite not always communicating clearly. They had high internal cohesion and low affirmation rates, often posting lengthy messages (.21 SDs above average) at the beginning of discussions (mean relative tree depth of .85 SDs below average). Their communication style

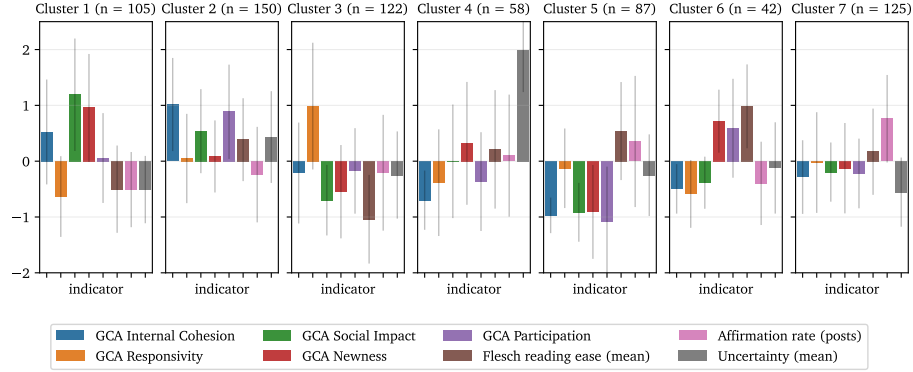


Figure 1: Participant Cluster Centroids (hand-selected features; SD as error bar)

was very certain, similar to cluster 6. They are comparable to the *Influential Actors* from Dowell, Nixon, and Graesser [10].

Cluster 2 students had outstanding participation rates. They communicated using simpler language and had a similar affirmation rate to cluster 3, while displaying some uncertainty in their communication style. Their social impact was high and their responsivity about average. This aligns them with the *Drivers* group from Dowell, Nixon, and Graesser [10]. Interestingly, they were also high performers on the cognitive activation exam assignment (.25 points; SD .76) and had the lowest dropout rate at 4% (as defined by if they attended the exam).

Cluster 3 members, dubbed *Summarizers*, displayed high responsivity and frequently rephrased their peers' information. They had low scores for newness and social impact, second only to cluster 5. Their participation tended to be late in the conversation (mean post depth: .57 SDs above average; SD 1.21), with less readable messages and a similar affirmation rate to cluster 2, but with little uncertainty. Cluster 3 often co-occurred with *Influential Actors* or *Drivers*, indicating an inter-dependency of these roles.

Cluster 4 stands out by having the most uncertain students while its members' posts have the second-highest newness on average. Members of this cluster don't participate very much and communicate with a low internal cohesion. Their social impact is near the average. These seem to be rather cautious students who are otherwise doing relatively average in most of the other measures. We refer to members of cluster 4 as *Cautious Learners*.

Cluster 5, or *Left-Behinds*, rarely participated in discussions and tended to join late. They didn't bother much to refer to their peers posts. Instead, they often only made affirmations accompanied by an excuse, giving them the second-highest affirmation rate. Their internal cohesion is to be taken with a grain of salt: this value drops if a participant does not make too many posts. *Left-Behinds* typically co-occurred with *Drivers* (co-occurrence in 27% of all groups; 78.16% of all cases where cluster 5 occurred). This also drove down their participation

score. They had the lowest average exam scores and the highest dropout rate at 25.3%, with most dropouts (19 out of 22) occurring in groups with *Drivers*.

In **cluster 6**, members participate actively and often contribute new information while writing readably, which is also their most defining characteristic. Their affirmation rate is slightly below average and thus among where the other more well-performing clusters are. While members of this cluster also performed well in the exam and had the second-lowest dropout rate, they did not perform very well on the social metrics of internal cohesion, responsivity or social impact, leading to the conclusion that these are people who don't communicate very effectively. This smallest cluster of only 42 students therefore seems to be most equivalent to what Dowell, Nixon, and Graesser [10] call *Chatterers*.

Cluster 7 members participated moderately, having high affirmation rates and typically contributed in the middle of the conversation, with a near average mean post depth. They waited until others had made their points, before affirming them and confidently adding their own information, with the lowest uncertainty among *Influential Actors*. Most of their scores were slightly below average, making them distinct from *Followers* from Dowell, Nixon, and Graesser [10], but more effective communicators than *Chatterers*. We call them *Deferers*.

The emergent roles that we found in this analysis are not only to be viewed on an individual level. Instead, by pairing the findings about the behavior of the cluster members with a co-occurrence matrix⁹, we could show that the behavioral patterns of one student produce those of another. Saqr and López-Pernas [28] propose that individual disposition influences emergent roles, yet group dynamics also steer students towards specific roles. When formulating feedback, it could severely demotivate students who took a less active or beneficial role in the face of, e.g., their peers being high performers that are already hard to keep up with. Therefore, all feedback should have a motivational component to it.

8 Feedback Design and Evaluation

We used our insights about the emergent roles and the indicators to create feedback for another cohort of students who worked on the same assignment. The feedback structure drew from Brooks *et al.* [2], who provide key questions for recalling learning goals (*feeding up*), reflecting on performance (*feeding back*), and providing tips for improvement (*feeding forward*). Brooks *et al.* [2] also identified four *levels* of feedback. First, the feedback should inform about a learners understanding of the task. For this, we developed a separate method to generate feedback on the product of phase 3. It draws from experiences we've made before [14] and is out of scope for the work at hand. Here, we focus on the *process*, the *self-regulation* and *self levels*. We gave students positively phrased feedback based on their role and helped them with self-regulation regarding their group behavior. As our method needs a fair amount of data and we didn't want students to try and manipulate indicators without us providing an interpretation, we sent them our feedback after the assignment finished.

⁹ You can find the co-occurrence matrix in the supplemental material.

Each emergent role got its own feedback template. The *feeding up* part – a short description of the assignment to recall the learning goal – was the same for all students. We then combined *feeding back* and *feeding forward* for indicators that actually mattered to the specific roles, as well as to the students in general. We guarded most of the clauses in the templates by conditionals. The condition for a clause to appear was usually a fixed deviation of the individuals from the z-score normalized mean value over the full sample, but sometimes also over a single phase. The conditionals and phrases were fine tuned so that no student would get an empty feedback text. We also added some clauses that addressed the co-occurrence of another role within the group, to address the emergence of certain roles in the light of another role being present.

Shortly after we designed the feedback, the new cohort of first semester teacher students finished the described CSEL assignment. Using the previous year’s cluster centroids, we executed our analysis code to obtain indicator values and emergent roles. For half the students, we generated the process feedback texts described here¹⁰ and combined this with the product feedback from our second feedback generation method. Students could receive either this *highly informative feedback* (HIF), or the *feedback as usual* (FAU), which was the same that the first cohort received. Besides the *feedup*, the FAU only showed basic information about the process. That included the *message count* for each phase and if they had participated enough so that the assignment could be counted as being solved. We included this information in the HIF, too. We now carried out a randomized experiment, where we presented the HIF to about half ($n = 202$) of the students and the FAU to the other half ($n = 206$).

We showed the resulting feedback texts to the students in our Learning Management System. To assert whether we should continue pursuing our course, we asked them to fill out a survey on the feedback, just below the feedback text. On a four point Likert-scale¹¹, we asked the students if the feedback (1) was comprehensible, (2) was helpful for their further learning, (3) made them aware of their progress, (4) helped them with reflecting their learning behavior, (5) helps them to self-regulate in the future, and (6) motivates them to learn. When compared to the FAU, all of these improved substantially. Not only that but the students liked our *highly-informative feedback* absolute terms: 79 % found the HIF to be helpful or very helpful (+28% to FAU; $U = 3236$; $p < 0.01\%$), two thirds found it helpful or very helpful for their reflection (+23%; $U = 3576.5$; $p = 0.01\%$) and 71 % replied that it would help them to self-regulate (+30%; $U = 3140$; $p < 0.01\%$). As this paper is about process feedback, these values are the most important ones here. In a Mann-Whitney U test, the differences between HIF and FAU of those three variables were highly significant ($p < 0.1\%$), showing that the students perceived the HIF to be an improvement over the FAU.

¹⁰ again, the feedback templates can be found in our OSF.io repository.

¹¹ (strongly) disagree, (strongly) agree

9 Conclusion and Outlook

We now have evidence that using linguistic indicators and GCA is a viable option to create data-informed, personalized, textual process feedback for learners (RQ1a). The analysis revealed emergent roles and gave us a glance at their interplay (RQ1b). It also revealed a path towards highly informative process feedback (RQ2a) that is based on a deeper language-based analysis of group dynamics, as well as individual writing styles. We need to give special care to the CSCL micro script when designing feedback based on such indicators and when analyzing emergent roles, as its influence is high for some indicators.

While our study was designed to give generated process feedback to all students alike, we think that there are some cases, like *Left Behinds*, where it would be better to redirect students to teachers instead. Adding to that, as the process feedback can be manipulable by students, our method should not be used to assess them summatively, but only to provide formative feedback.

Our preliminary survey results indicate that we're making progress towards providing useful automated feedback (RQ2b) through our presented method. Jivet *et al.* [18] demanded that LA-informed feedback would need to be contextualized in order to be relevant to students. They demanded this in the light of LA dashboards which back then were often missing the context needed to make sense of the presented indicators. Our concise, actionable, and contextualized textual feedback meets this demand. Of course, the students' self-reported value judgments have their limitations. We now have to test whether the students actually adapt their behavior in response to the feedback. This could show in improvements in single indicators as well as in the role they take in a discussion.

Again, we ran assignments with one cohort of students so that we could find indicators and create models that would be helpful to create feedback for another cohort doing the same assignment. This advances on prior methods for LA-informed feedback and sets the path for our future research trajectory. Regarding the process feedback, using our proposed method, we will need to find more indicators that are stable across different micro scripts and create sets of indicators and their empirical values that are specific to certain micro scripts as well. This will bring us towards process feedback that is not only useful for a certain assignment macro script, but which will also be composable and reusable across different assignments.

Lowering the integration barrier of automated feedback into teaching practice remains a challenge. Under our current method, LA experts must laboriously identify process indicators, contextualize them and phrase feedback accordingly. To scale this up, we will need a technical infrastructure as well as processes in place that support us in collecting empirical values of experience on the indicators. The results must also be of obvious value to teachers. In the long run, we hence need to support them in the authoring process. Integrated into tools that provide teachers information on their students performance [e.g., 19], we should deliver recommendations of indicators and role-specific feedback texts, that are tailored to their assignments automatically.

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